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EXSUM on VTC (7MAY20 -1200-1300 hrs) with Dr. Scott Gottlieb, EAB Member

Meeting was led by COO Andy Makridis and arranged by [REDACTED]
Representatives from OGI and WCPMC attended.

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Dr. Gottlieb primarily provided his thoughts on the state of US based medical countermeasures/prophylaxis development arranged into 3 broad types: mRNA based protein expression, DNA adenovirus shell vaccine delivery and traditional recombinant protein vaccines. He answered a broad range of questions regarding the current clinical manifestations of COVID-19, implications of seasonality, state of US seroprevalence, asymptomatic infected individuals and comparisons to foreign countries' COVID-19 responses. The biggest takeaway is that he believes the PRC will be first to market with a vaccine against COVID-19 and is concerned that this will have positive geopolitical repercussions for the PRC even if the product proves to be marginally effective.

Speaker Bio

Dr. Scott Gottlieb is an American physician and investor who served as the 23rd commissioner of the FDA from 2017 until April 2019. In April 2020 he joined Maryland Governor Larry Hogan's COVID-19 Response Team. He also advises Massachusetts Governor Charlie Baker on COVID-19 and serves as an Advisor on the Reopen Connecticut Advisory Group. Gottlieb is presently a resident fellow at the American Enterprise Institute (AEI), a partner at the venture capital firm New Enterprise Associates, a member of the Board of Directors of Pfizer and a contributor to the cable financial news network CNBC. Before becoming FDA commissioner, he was a clinical assistant professor at New York University School of Medicine, the FDA deputy commissioner for medical and scientific affairs, a venture partner with New Enterprise Associates, a member of the policy board of the Leukemia & Lymphoma Society, and a senior official at the Centers for Medicare and Medicaid Services.

Selected Details

Dr. Gottlieb believes that cases will continue to rise in the US through May based on a R_0 (BRR) value of 3 for SARS-CoV-2. Current testing is estimated to be detecting between 1 in 10 and 1 in 20 infections, suggesting that the US could experience 300,000 cases.

The trends also suggest a seasonality to COVID-19 waning with the warmer months in the US, similar to H1N1, but returning in the fall season. He is closely following cases as Brazil enters their winter season to determine if COVID-19 will track as a seasonal respiratory viral infection.

He mentioned several treatment modalities all as marginal or ineffective. He sees the most promise in Remdesivir as the most promising (although also of moderate efficacy) treatment option due to Regeneron's complete in-house capability to produce the drug from lab to industrial scale. Other manufacturers outsource significant portions of scale production.

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He firmly believes the US must have domestic production of medical countermeasures as prior epidemics led to predictable nationalization of such resources in other countries. Dr. Gottlieb described 3 broad methods for vaccine development a) mRNA based using exogenous expression of the S (or spike) protein independently being pursued by Pfizer & Moderna, b) adenovirus based DNA based vaccine delivery pursued by Johnson & Johnson, Oxford, & Astra-Zeneca and c) recombinant protein vaccine pursued by Innova, Sanofi/GlaxoSmithKline. This last approach seems to require an adjuvant as the protein being expressed is apparently not very immunogenic.

Dr. Gottlieb has not been following the international medical countermeasures field as closely as US ones, but believes that the PRC has several candidates in clinical trials and that they will be the first to market what is likely to be a substandard vaccine. He is concerned that being "first" has prestige (even if relatively ineffective) and will be used to advantage by the PRC geopolitically.

Queried about the experience other countries are having, his general impression is that smaller countries such as the Republic of Korea, the governing authorities on Taiwan, the semi-autonomous region of Hong Kong and the Republic of Singapore were all successful as they could impose restrictions on civil liberties that would not be acceptable to US citizens. Comparing Stockholm to NYC, he sees that the rate and scale of infections is about the same (from a Journal of the American Medical Association manuscript published this week by the Karolinska Institute), so the jury is out on whether locking down (or not) was effective; he believes Stockholm did effectively lockdown even if not declared.

On asymptomatic carriers, nasopharyngeal swabs have been obtained from patients from which virions were recovered and suggest between 30 – 60% of infected individuals could be shedding virus with no apparent symptoms. These individuals appear to be mounting an immune response from seroprevalence studies with an age dependence implying that SARS-CoV-2 may not be strongly immunogenic.

On "herd immunity" Dr. Gottlieb discounts the 2 major studies from Los Angeles and Santa Clara counties showing high seroprevalence. He believes nationally, the seroprevalence is about 20-25% which is insufficient for herd immunity at the R_0 (BRR) value of 3.

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